

Course Code	
Course Name	Angular

Duration (in days)	8	Proficiency Level	Front End Development
Pre-requisites	None	Target Audience	Campus Hires

Learning Outcome

At the end of the program, participants will be able to learn:

- Understand Single Page Applications and their benefits.
- Awareness of various Javascript libraries / frameworks for designing Single Page Applications.
- Ability to consume NPM packages and use Angular CLI to setup dev environment.
- Knowledge of Angular Bootstrapping.
- Debug Angular code using debugging tools.
- Develop Angular Applications using the building blocks like Modules, Components, Pipes, Directives and Decorators
- Establish Inter-component communication.
- Understand Meta Data and Data Binding techniques.
- Awareness of Built-in Pipes and Directives
- Implement Services & Dependency Injection.
- Ability to integrate bootstrap with Angular for rich user experiences.
- Implement Lifecycle Hooks and describe various life cycle events of components.
- Good working knowledge in Creating and validating Template and Reactive Forms.
- Consume external web services / RESTful services using RXJS.
- Implement Routing by creating Route Definition Object, Router-Outlets.

Day-wise Session Plan

Day	Unit	Objective(s)	Hours
1	Getting Started	• What are Single Page Applications (SPA) • Characteristics of Single Page Applications • JavaScript Frameworks used to design and develop SPA.	2
1	Angular Introduction	• What is Angular? • Angular Environment and IDEs • Installing Angular CLI and Creating a sample project • Directory structure and various configuration files • Bootstrapping Angular Application • Understanding the Role of AppModule and Component Declaration • Angular CLI commands • Installing Bootstrap and styling your application • Decorators in Angular	6
2	Components and DataBinding	• What are Components? • Manually Creating Components • Working with Component Templates and Component Styles • Data Databinding • String Interpolation	4

		- Property Binding - One Way Binding - Two Way Binding - Event Binding - Passing Data between Components	
2, 3	Directives	- Understanding Directives - Structural Directives and Attribute Directives - ngIf to Output Data Conditionally - ngIf with an Else Condition - Styling Elements Dynamically with ngStyle - CSS Classes Dynamically with ngClass - Outputting Lists with ngFor - Getting the Index when using ngFor - Understanding ngSwitch - HostListener to Listen to Host Events - HostBinding to Bind to Host Properties	6 (4+2)
3	Debugging and Error handling	- Understanding Angular Error Messages - Using VS Code to debug Angular Applications - Using Angular DevTools to Dive into Angular Apps - Creating and using Global Custom ErrorHandlers - Using the new ng Object	4
3, 4	Life Cycle Hooks	- Understanding the Component Lifecycle - Seeing Lifecycle Hooks in Action - Tracking Changes - Lifecycle Hooks and Template Access @ViewChild() - Getting Access to the Template & DOM with @ViewChild - Projecting Content into Components with ng-content @ContentChild() decorator - Getting Access to ng-content with @ContentChild	4 (2+2)
4	Pipes	- Pure and Impure Pipes - Using Built in Pipes - The PipeTransform Interface - Creating Custom Pipes - Performance implications of Impure Pipes	6
5	Using Services & Dependency Injection	- Understanding DI in Angular - Constructor based hierarchical Injector - Singleton and non-Singleton Objects - Using Service Classes - Working with various Decorators in DI @Injector() @Injectable()	8
6	Forms Programming	- Building Template Forms and Reactive Forms - Understanding FormBuilder, FormGroup, FormControl classes - Validating Forms - Creating Custom Validators - Implementing Asynchronous Validations - Submitting form data to the server	8
7	RXJS and Http Programming	- RXJS Introduction - Observables, Observers and Subjects	8

		• Consuming Web Services • Passing parameters using HTTP Headers • Creating custom Http Interceptors • Error handling in RXJS	
8	Routing	• Routing Introduction • Route Definition Object • Router Outlets • Passing Parameters to Routes • Life cycle of Router • Implementing Guards and Resolvers	8